METAL FORMING

Problem 1 (06/11/2020)

A cylindrical sample in 6061-O aluminum alloy (k = 205 MPa, n = 0,2) with an initial height $h_0 = 50$ mm and an initial diameter $d_0 = 40$ mm, is compressed between two flat dies.

1) Determine the diameter of the sample if the true strain is equal to $\varepsilon = -0{,}16$.
2) The process requires two steps: a first one to a height $h_1 = 43$ mm, followed by a second step to a height $h_2 = 37$ mm. Neglecting friction, determine the force required for each step, and the ideal work spent on the overall deformation.
3) In the hypothesis that the total strain energy density to reach the final shape is equal to $u = 0{,}15$ J/mm^3, determine the cost of the energy required to deform 5000 samples (cost of the energy equal to $0{,}1$ €/kWh).

Solution

1) Determine the diameter of the sample if the true strain is equal to $\varepsilon = -0{,}16$.

Considering the volume as constant:

$$V_0 = \frac{\pi d_0^2}{4} h_0 = V_f = \frac{\pi d_f^2}{4} h_f$$

Being the final height equal to:

$$\varepsilon = \ln\left(\frac{h_f}{h_0}\right)$$

$$h_f = h_0 e^{\varepsilon} = 50 \cdot e^{-0{,}16} = 42{,}61 \text{ mm}$$

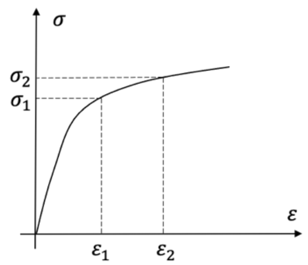
The final diameter of the sample is equal to:

$$d_f = \sqrt{\frac{h_0}{h_f} d_0^2} = \sqrt{\frac{50}{42{,}61} 40^2} = 43{,}33 \text{ mm}$$

2) The process requires two steps: a first one to a height $h_1 = 43$ mm, followed by a second step to a height $h_2 = 37$ mm. Neglecting friction, determine the force required for each step, and the ideal work spent on the overall deformation.

Referring to the figure, the required forces are:

$$F_1 = \sigma_1 A_1$$
$$F_2 = \sigma_2 A_2$$



Where, the flow stresses are:

$$\sigma_1 = k\varepsilon_1^n$$
$$\sigma_2 = k\varepsilon_2^n$$

Being:

$$\varepsilon_1 = \ln\left(\frac{h_1}{h_0}\right) = \ln\left(\frac{43}{50}\right) = -0{,}15$$

$$\varepsilon_2 = \ln\left(\frac{h_2}{h_0}\right) = \ln\left(\frac{37}{50}\right) = -0{,}30$$

$$\sigma_1 = k\varepsilon_1^n = -205 \cdot 0{,}15^{0{,}2} = -140{,}27 \text{ MPa}$$

$$\sigma_2 = k\varepsilon_2^n = -205 \cdot 0{,}30^{0{,}2} = -161{,}13 \text{ MPa}$$

and the cross section of the specimen in the two steps equal to:

$$V_0 = \frac{\pi d_0^2}{4} h_0 = \frac{\pi 40^2}{4} 50 = 62.832 \text{ mm}^3$$

$$A_1 = \frac{V_0}{h_1} = \frac{62.832}{43} = 1.461 \text{ mm}^2$$

$$A_2 = \frac{V_0}{h_2} = \frac{62.832}{37} = 1.698 \text{ mm}^2$$

The required forces are:

$$F_1 = \sigma_1 A_1 = -140{,}27 \cdot 1.461 = -204.934 \text{ N} = -204{,}9 \text{ kN}$$

$$F_2 = \sigma_2 A_2 = -161{,}13 \cdot 1.698 = -273.599 \text{ N} = -273{,}6 \text{ kN}$$

The overall strain energy density is equal to:

$$u = \left|\frac{k\varepsilon_2^{n+1}}{n+1}\right| = \frac{205 \cdot 0{,}30^{0{,}2+1}}{0{,}2+1} = 40{,}3 \text{ MPa} = 0{,}0403 \ \frac{\text{J}}{\text{mm}^3}$$

Therefore, the ideal work spent on the overall deformation is equal to:

$$W = uV_0 = 0{,}0403 \cdot 62.832 = 2.532 \text{ J} = 2{,}5 \text{ kJ}$$

3) *In the hypothesis that the total strain energy density to reach the final shape is equal to $u = 0{,}15$ J/mm^3, determine the cost of the energy required to deform 5000 samples (cost of the energy equal to $0{,}1$ €/kWh).*

The ideal work to produce a single sample is equal to:

$$W = uV_0 = 0{,}15 \cdot 62.832 = 9.425 \text{ J}$$

The ideal work to produce 5.000 samples is equal to (recall $1 \text{ kWh} = 3{,}6 \text{ MJ}$):

$$W_{total} = nW = 5.000 \cdot 9.425 = 47.125.000 \text{ J} = 47{,}125 \text{ MJ} = \frac{47{,}125}{3{,}6} \text{kWh} = 13{,}1 \text{ kWh}$$

Therefore, the cost of energy is equal to:

$$C_E = W_{total} \cdot c_{ele} = 13{,}1 \text{ kWh} \cdot 0{,}1 \frac{€}{\text{kWh}} = 1{,}31 \text{ €}$$

Problem 2 (15/01/2021)

The height of an annealed carbon steel cylinder with a diameter equal to 50 mm and a height equal to 100 mm undergoes to open-die forging. The material is characterized by a flow curve with parameters k = 500 MPa and n = 0,24, and UTS equal to 350 MPa. The process requires two steps in both of which the height is reduced by 5%. Neglecting friction and barreling phenomenon, determine:

1) The strains for each of the two steps and the overall strain.
2) The ideal work for each of the two steps.

Solution

1) Determine the strains for each of the two steps and the overall strain

The process requires two steps in both of which the height is reduced by 5%, therefore the height of the cylinder will become, respectively:

Step 1: $h_1 = h_0 \cdot (1 - 0{,}05) = 100 \cdot 0{,}95 = 95{,}00\ \mathrm{mm}$

Step 2: $h_2 = h_1 \cdot (1 - 0{,}05) = 95 \cdot 0{,}95 = 90{,}25\ \mathrm{mm}$

Therefore, the true strains are:

Step 1: $\varepsilon_1 = \varepsilon_{0-1} = \ln\frac{h_1}{h_0} = \ln\frac{95}{100} = -0{,}0513$

Step 2: $\varepsilon_{1-2} = \ln\frac{h_2}{h_1} = \ln\frac{90{,}25}{95} = -0{,}0513$

Overall: $\varepsilon_2 = \ln\frac{h_2}{h_0} = \ln\frac{90{,}25}{100} = -0{,}1026 = \varepsilon_{0-1} + \varepsilon_{1-2}$

2) Determine the ideal work for each of the two steps

Please, recall that the strain energy density when dealing with the considered flow curve is equal to $u = \left|\frac{k \cdot \varepsilon^{n+1}}{n+1}\right|$, therefore:

Step 1: $u_{0-1} = \left|\frac{k \cdot \varepsilon_1^{n+1}}{n+1}\right| = \frac{500 \cdot 0{,}0513^{0{,}24+1}}{0{,}24+1} = 10{,}141\ \mathrm{MPa} = 0{,}010141\ \frac{\mathrm{J}}{\mathrm{mm}^3}$

Overall: $u_{0-2} = \left|\frac{k \cdot \varepsilon_2^{n+1}}{n+1}\right| = \frac{500 \cdot 0{,}1026^{0{,}24+1}}{0{,}24+1} = 23{,}954\ \mathrm{MPa} = 0{,}023954\ \frac{\mathrm{J}}{\mathrm{mm}^3}$

Step 2: $u_{1-2} = u_{0-2} - u_{0-1} = 0{,}023954 - 0{,}010141 = 0{,}013813\ \frac{\mathrm{J}}{\mathrm{mm}^3}$

Now, considering the volume of the cylinder:

$$V_0 = \frac{\pi}{4} d_o^2 \cdot h_0 = \frac{\pi}{4} 50^2 \cdot 100 = 196.350\ \mathrm{mm}^3$$

The ideal work for each of the two steps will be:

Step 1: $W_{0-1} = u_{0-1} \cdot V_0 = 0{,}010141 \cdot 196.350 = 1.991\ \mathrm{J} \cong 2\ \mathrm{kJ}$

Step 2: $W_{1-2} = u_{1-2} \cdot V_0 = 0{,}013813 \cdot 196.350 = 2.712\ \mathrm{J} \cong 2{,}7\ \mathrm{kJ}$

Problem 3 (21/06/2021)

A cylindrical sample, with an initial height $h_0 = 500$ mm and an initial diameter $d_0 = 400$ mm, is hot forged between two flat dies. The material behaves like a perfectly plastic material with $Y = 320\ \text{MPa}$.

1) Neglecting friction, determine the ideal work required to reach a true strain equal to $\varepsilon = -0{,}05$.
2) Neglecting friction, determine the final height of the sample if the true strain is equal to $\varepsilon = -0{,}06$.
3) Neglecting friction, determine the maximum sample diameter that can be obtained with a press capable of achieving a maximum force equal to 50 MN (a 50 MN press).
4) Assuming that the friction is not negligible and is equal to $\mu = 0{,}15$, determine the force to obtain a final diameter equal to $d_1 = 430\ \text{mm}$. Check if the open die forging operation can be done with a 65 MN press.

Solution

1) Neglecting friction, determine the ideal work required to reach a true strain equal to $\varepsilon = -0{,}05$.

The ideal work required is equal to:

$$W = uV_0$$

where the volume of the cylindrical sample V_0 is:

$$V_0 = \frac{\pi d_0^2}{4} h_0 = \frac{\pi \cdot 400^2}{4} \cdot 500 = 62.831.853\ \text{mm}^3$$

and u is the strain energy density

$$u = |Y\epsilon| = 320 \cdot 0{,}05 = 16\ \text{MPa} = 0{,}016 \frac{\text{J}}{\text{mm}^3}$$

It follows

$$W = 0{,}016 \cdot\ 62.831.853 = 1.005.310\ \text{J} = 1{,}01\ \text{MJ}$$

2) Neglecting friction, determine the final height of the sample if the true strain is equal to $\varepsilon = -0{,}06$.

The material behaves like a perfectly plastic material, therefore there is no elastic recovery. It follows that the final true strain is $\varepsilon = -0.06$.

Recalling that

$$\varepsilon = \ln \frac{h_1}{h_0}$$

The final height of the sample is

$$h_1 = h_0 e^{\varepsilon} = 500\ e^{-0{,}06} = 470{,}88\ mm$$

3) Neglecting friction, determine the maximum sample diameter that can be obtained with a press capable of achieving a maximum force equal to 50 MN (a 50 MN press).

The maximum force is equal to $F_{max} = Y \cdot A_{max} = Y \cdot \frac{\pi}{4} d_{max}^2$

Therefore, the maximum diameter is equal to

$$d_{max} = \sqrt{\frac{4 \cdot F_{max}}{\pi \cdot Y}} = \sqrt{\frac{4 \cdot\ 50.000.000}{\pi \cdot 320}} = 446{,}031\ \text{mm}$$

4) Assuming that the friction is not negligible and is equal to $\mu = 0{,}15$, determine the force to obtain a final diameter equal to $d_1 = 430$ mm. Check if the open die forging operation can be done with a 65 MN press.

The force can be computed as

$$F = p_{av} \frac{\pi d_1^2}{4} = Y\left(1 + \frac{2\mu r_1}{3h_1}\right) \cdot \frac{\pi d_1^2}{4}$$

where h_1 can be computed considering the volume of the sample constant

$$h_1 = \frac{d_0^2}{d_1^2} h_0 = \frac{400^2}{430^2} \cdot 500 = 432{,}666\ \text{mm}$$

Ne segue:

$$F = 320\left(1 + \frac{0{,}15 \cdot 430}{3 \cdot 432{,}666}\right) \cdot \frac{\pi}{4} 430^2 = 48.779.643\ \text{N} = 48{,}8\ \text{MN} < 65\ \text{MN}$$

The forging operation can be done on a 65 MN press.